Prompts

Data Selection

You are a highly skilled data scientist tasked with selecting the most appropriate dataset for a specific project. Your task requires careful consideration of the dataset's properties, its compatibility with the intended application, and its ability to meet predefined criteria.

Objective of the Dataset Selection Task:
<Input Dataset Details> <Task>

Available Dataset Options
Dataset 1: AudioCommandLite
- **Description:** A small, labeled dataset of short voice commands for classification tasks.
- **Key Features:**
  - Modality: Audio
  - Samples: 3000
  - Classes: 10
  - Annotation Quality: High
  - Noise Level: Low

Dataset 2:
<Details>
Criteria for Dataset Selection
- Modality: Audio
- Max Samples: <= 5000
- Noise Robustness: Yes
- Classes: At least 10
- Annotation Quality: High

Instructions for Selection
1. **Review the Dataset Options:** Examine each dataset’s description and features.
2. **Compare with Desired Features:** Match the dataset’s attributes against the listed criteria.
3. **Evaluate Additional Factors:** Consider dataset size, diversity, and metadata.
4. **Make a Justified Decision:** Select the dataset that best aligns with the project objectives.

Summary
Analyze datasets carefully and justify your selection using clear reasoning and evidence-based comparison.

Model Selection

You are a highly skilled and detail-oriented machine learning engineer. Your expertise is required to select the most suitable model for a specific task. You will evaluate several candidate models based on their performance metrics, behavior, and alignment with the given criteria to make an informed and well-justified decision.
Task Description
The model selection is for the following task:
<User Dataset Details>
Dataset Description
The task uses the dataset described as follows:
<Toy / Synthetic Dataset Details>
The table below shows the column names, their value formats, and their description.
Available Models and Their Performance Metrics
Below are details of the candidate models. Each model has been evaluated based on its performance metrics across training and validation phases. Carefully review the details to identify patterns, strengths, and weaknesses.
Model 1: GRUClassifier
- **Summary:** A classifier using a Gated Recurrent Unit (GRU) backbone. GRUs are a variant of RNNs that can capture long-term dependencies in sequential data, offering better performance than vanilla RNNs.
- **Performance Metrics:**
- Loss Significance: 0.2208861304796224
- Validation Loss Significance: None
- Epoch-wise Loss: e1:0.3793621957302093; e2:0.337876409292221; e3:0.2444266378879547; e4:0.1732485294342041; e5:0.137539729475975
- Epoch-wise Validation Loss: e1:0.5928476452827454; e2:0.4688830971717834; e3:0.2080093622207641; e4:0.1987400352954864; e5:0.2091681659221649
- Mean Absolute Error (MAE): e1:0.4840775728225708; e2:0.4537611305713653; e3:0.3829525709152221; e4:0.3242017030715942; e5:0.288499653339386
- Mean Absolute Percentage Error (MAPE): e1:762.216796875; e2:806.5173950195312; e3:735.75439453125; e4:631.6510009765625; e5:584.645263671875
- Validation MAE: e1:0.637512743473053; e2:0.5608084201812744; e3:0.3549753129482269; e4:0.3491454124450683; e5:0.3600536286830902
- Validation MAPE: e1:804.3636474609375; e2:678.9108276367188; e3:624.7390747070312; e4:662.1983032226562; e5:628.3173828125
Model 2: MLPClassifier
- **Summary:** A multi-layer perceptron (MLP) classifier with a fully connected network

backbone. It is commonly used for tasks where the data does not have a sequential or temporal structure.
- **Performance Metrics:**
<Metrics for the model>
Model 3: LSTMClassifier
- **Summary:** A classifier using a Long Short-Term Memory (LSTM) backbone. LSTMs are a type of RNN that can capture long-term dependencies and handle vanishing gradient problems in deep sequences.
- **Performance Metrics:**
<Metrics for the model>
Model 4: FNetClassifier
- **Summary:** Uses an FNet backbone for processing input data. FNet employs a Fourier transformation in place of attention mechanisms, offering a more efficient way to model relationships between inputs.
- **Performance Metrics:**
<Metrics for the model>

Model 5: gMLPClassifier
- **Summary:** A classifier using the gMLP (Gated MLP) backbone, which is a type of multi-layer perceptron designed to model long-range dependencies using gated transformations.
- **Performance Metrics:**
<Metrics for the model>
Model 6: MixerClassifier
- **Summary:** Utilizes a Mixer backbone, which employs a permutation-equivariant model for processing data by mixing different channels and spatial dimensions.
- **Performance Metrics:**
<Metrics for the model>
Model 7: TransformerClassifier
- **Summary:** A transformer-based classifier, utilizing multi-head attention mechanisms for capturing relationships across inputs, particularly useful in sequence-to-sequence tasks like text or time-series classification.
- **Performance Metrics:**
<Metrics for the model>
Model 8: RNNClassifier
- **Summary:**A recurrent neural network classifier using an RNN backbone for sequential

data processing. It can handle sequential inputs and is efficient for tasks like time-series and text data classification
- **Performance Metrics:**
<Metrics for the model>
Model 9: ExternalTransformerClassifier (1)
- **Summary:** A transformer classifier using an external transformer backbone with attention mechanisms designed for large-scale or external data processing.
- **Performance Metrics:**
<Metrics for the model>
Model 10: ConvMixerClassifier (1)
- **Summary:** Uses a ConvMixer backbone, combining convolutional layers with mixers for efficiently processing high-dimensional input data, typically for image-based classification tasks.
- **Performance Metrics:**
<Metrics for the model>

Model 11: ParNetAttentionClassifier (1)
- **Summary:** A classifier using the ParNet (Position-Aware Relation Network) backbone, which focuses on improving image-text matching by modeling both semantic and spatial relationships.
- **Performance Metrics:**
<Metrics for the model>
Model 12: ConvMixerClassifier
- **Summary:** Uses a ConvMixer backbone, combining convolutional layers with mixers for efficiently processing high-dimensional input data, typically for image-based classification tasks.
- **Performance Metrics:**
<Metrics for the model>
Model 13: SEAttentionClassifier
- **Summary:** A classifier utilizing the SEAttention (Squeeze-and-Excitation) mechanism, which adaptively recalibrates channel-wise feature responses to improve performance in image classification tasks.
- **Performance Metrics:**
<Metrics for the model>
Model 14: ECAAttentionClassifier
- **Summary:** Uses an ECA (Efficient Channel Attention) backbone, providing efficient

channel attention via a 1D convolution mechanism to improve performance without significant model complexity.
- **Performance Metrics:**
<Metrics for the model>
Model 15: ParNetAttentionClassifier
- **Summary:** A classifier using the ParNet (Position-Aware Relation Network) backbone, which focuses on improving image-text matching by modeling both semantic and spatial relationships.
- **Performance Metrics:**
<Metrics for the model>
Model 16: PerformerAttentionClassifier
- **Summary:** Uses a Performer backbone with linear time and space complexity, leveraging a novel attention mechanism (FAVOR+) to approximate softmax attention, providing scalability in large tasks.
- **Performance Metrics:**
<Metrics for the model>
Model 17: CBAMClassifier
- **Summary:** A classifier using the CBAM (Convolutional Block Attention Module) backbone, which combines channel and spatial attention to improve the quality of feature extraction in image classification tasks.
- **Performance Metrics:**
<Metrics for the model>
Model 18: UFOAttentionClassifier
- **Summary:** A classifier using the UFO (Universal Feature-Oriented Attention) backbone, which focuses on attention mechanisms with flexible heads and dropout regularization to enhance performance.
- **Performance Metrics:**
<Metrics for the model>
Model 19: SimAMClassifier
- **Summary:** A classifier using the SimAM (Simple Attention Mechanism) backbone, focusing on a simple yet effective attention mechanism with minimal computational overhead.
- **Performance Metrics:**
<Metrics for the model>
Model 20: SwitchTransformerClassifier
- **Summary:** A classifier using a Switch Transformer backbone, which incorporates a mixture of experts approach for more efficient computation, using only a subset of model

parameters at each step.
- **Performance Metrics:**
<Metrics for the model>
Model 21: ExternalTransformerClassifier
- **Summary:** A transformer classifier using an external transformer backbone with attention mechanisms designed for large-scale or external data processing.
- **Performance Metrics:**
<Metrics for the model>